Insert your title here

Insert a subtitle here (if applicable)



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Submitted to the Department of Electrical Engineering at the University of Cape Town in partial fulfilment of the academic requirements for a Bachelor of Science degree in ***INSERT DEGREE NAME HERE (Electrical Engineering; Mechatronics or Electrical and Computer Engineering)

August 3, 2026

Declaration

1. I know that plagiarism is wrong. Plagiarism is to use another's work and pretend that it is one's own.
2. I have used the IEEE convention for citation and referencing. Each contribution to, and quotation in, this report from the work(s) of other people has been attributed, and has been cited and referenced.
3. This report is my own work.
4. I have not allowed, and will not allow, anyone to copy my work with the intention of passing it off as their own work or part thereof.

Signature:.....
M. S. Tšoeu

Date:.....

Terms of Reference

The terms of reference page is an agreement between yourself and your supervisor outlining what is expected of you in your final year project. Please make sure that this is discussed and written at the beginning of your thesis project.

Acknowledgments

Make relevant acknowledgements to people who have helped you complete or conduct this work, including sponsors or research funders.

Abstract

- Open the **Project Report Template.tex** file and carefully follow the comments (starting with %).
- Process the file with **pdflatex**, using other processors may need you to change some features such as graphics types.
- Note the files included in the **Project Report Template.tex** (with the .tex extension excluded). You can open these files separately and modify their contents or create new ones.
- Contact the latex namual for more features in your document such as equations, subfigures, footnotes, subscripts & superscripts, special characters etc.
- I recommend using the **kile** latex IDE, as it is simple to use.

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Chapter 1

Introduction

1.1 Background to the study

A very brief background to your area of research. Start off with a general introduction to the area and then narrow it down to your focus area. Used to set the scene [**?**].

1.2 Objectives of this study

1.2.1 Problems to be investigated

Description of the main questions to be investigated in this study.

1.2.2 Purpose of the study

Give the significance of investigating these problems. It must be obvious why you are doing this study and why it is relevant.

1.3 Scope and Limitations

Scope indicates to the reader what has and has not been included in the study. Limitations tell the reader what factors influenced the study such as sample size, time etc. It is not a section for excuses as to why your project may or may not have worked.

1.4 Plan of development

Here you tell the reader how your report has been organised and what is included in each chapter.

I recommend that you write this section last. You can then tailor it to your report.

Chapter 2

Literature Review

2.1 Tinkerkit Braccio arm for Arduno

The Braccio arm has been used in similar projects to demonstrate teleoperation via intuitive control (natural human motions). An example uses an armband to measure the muscle contractions of a human arm [1].

The Braccio arm has also been used in other projects. For a feeding-assistant, the engineers used Denavit-Hartenberg parameters and forward-kinematics to determine the position of the end-effector [2]. The Denavit-Hartenberg parameters used in the study are shown in Table 2.1.

Table 2.1: Denavit-Hartenberg parameters for Braccio arm

i	α_{i-1} **(degrees)**	r_{i-1} **(inches)**	d_i **(inches)**	θ_i (degrees)
1	0	0	0	θ_1
2	90	0	0	θ_2
3	0	5	0	θ_3
4	0	5	0	θ_4
5	0	12	0	θ_5

There was a project involving the use of computer vision to control the Braccio arm [3]. In this project, a camera connected to a Raspberry Pi would detect hand gestures (number of fingers being held up) from the user, and translate these into commands for the Braccio arm.

2.2 Teleoperation

Teleoperation has generally been in three categories [4]:

- Joint mapping: Each joint of the slave device is mapped to a joint of the master device. This is only applicable if the kinematics are similar, which is not the case for a human arm and the Braccio arm.
- End-effector mapping: The end-effector of the slave device is mapped to the end-effector of the master, and the intermediate joints are determined with forward and inverse kinematics.
- Pose mapping: The general intention of the master device is mapped to the slave device.

The paper found that pose mapping, using a linear transformation in an intermediate space, was not only simpler to implement, but also more intuitive to use. The measure of "intuitiveness" comes from how quickly a novice was able to complete a task.

2.3 Computer Vision

The works of Moeslund and Granum [5] summarises the advancements of computer vision in human tracking, from 1980 up till halfway through year 2000. The paper classifies human tracking into three applications; surveillance, control and analysis. From the descriptions given, control is what's relevant to this project.

The paper lists some common assumptions made in the context of computer vision. Computer vision isn't the focus of this project, so as many assumptions as applicable will be made, namely:

- The subject (human arm) remains inside the workspace
- No camera motion
- Only one person in the workspace at the time
- No occlusion

- The motion pattern of the subject is known
- Minimal changes to lighting
- Static/uniform background
- Known camera parameters

That being the case, robustness of a tracking algorithm is not as much of a priority, compared to accuracy and speed.

The paper also discusses different ways of modelling the hand, from different abstraction methods. The stick-figure model, with joint-abstraction stands out, because it can most easily be translated into a corresponding pose for the Braccio arm.

Chapter 3

Methodology

This is what I did to test and confirm my hypothesis.

You may want to split this chapter into sub chapters depending on your design. I suggest you change the title to something more specific to your project.

This is where you describe your design process in detail, from component/device selection to actual design implementation, to how you tested your system. Remember detail is important in technical writing. Do not just write I used a computer give the computer specifications or the oscilloscopes part number. Describe the system in enough detail so that someone else can replicate your design as well as your testing methodology.

If you use or design code for your system, represent it as flow diagrams in text.

IMPORTANT: Include a motivation for your selection of an appropriate technique, or engineering or IT tool to solve your project problem. Discuss any limitations if appropriate.

Chapter 4

Results

These are the results I found from my investigation.

Present your results in a suitable format using tables and graphs where necessary. Remember to refer to them in text and caption them properly.

4.1 Simulation Results

4.2 Experimental Results

Chapter 5

Discussion

Here is what the results mean and how they tie to existing literature...

Discuss the relevance of your results and how they fit into the theoretical work you described in your literature review.

Chapter 6

Conclusions

These are the conclusions from the investivation and how the investigation changes things in this field or contributes to current knowledge...

Draw suitable and intelligent conclusions from your results and subsequent discussion.

Chapter 7

Recommendations

Make sensible recommendations for further work.

Bibliography

[1] A. Mongardi, F. Rossi, A. Prestia, D. Demarchi, and P. M. Ros, "Live demonstration: A wearable armband for real-time control of multi-dof robotic actuators," in *2023 IEEE International Symposium on Circuits and Systems (ISCAS)*, 2023, pp. 1–1. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/10181982

[2] M. E. Emu, S. Biswas, and R. Shrestha, "Development of a feeding assistive robot using a six degree of freedom robotic arm," 2023. [Online]. Available: https://arxiv.org/abs/2309.11594

[3] A. Reyes, S. Reinhardt, T. Wise, N. Rawashdeh, and S. Paheding, "Gesture controlled collaborative robot arm and lab kit," in *2022 CIEC*, 2024. [Online]. Available: https://peer.asee.org/45693.pdf

[4] C. Meeker, T. Rasmussen, and M. Ciocarlie, "Intuitive hand teleoperation by novice operators using a continuous teleoperation subspace," in *2018 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2018, pp. 5821–5827. [Online]. Available: https://ieeexplore.ieee.org/abstract/document/8460506

[5] T. B. Moeslund and E. Granum, "A survey of computer vision-based human motion capture," *Computer Vision and Image Understanding*, vol. 81, no. 3, pp. 231–268, 2001. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S107731420090897X

Appendix A

Additional Files and Schematics

Add any information here that you would like to have in your project but is not necessary in the main text. Remember to refer to it in the main text. Separate your appendices based on what they are for example. Equation derivations in Appendix A and code in Appendix B etc.

IMPORTANT: Appendix A (see the table below) should provide a summary of how you have met the GAs associated with the course, and where in the report the evidence can be found.

If you have used AI in writing your report, you need to provide details in one of the appendices of how you used it.

If appropriate, in a subsequent appendix, provide a link to any GitHub repository where the code and additional materials for your project can be found.

GA	Requirement	Justification and section in the report
1	Problem-solving	Description here
4	Investigations, experiments and data analysis	Description here
5	Use of engineering tools	Description here
6	Professional and technical communication (Long report)	Description here
8	Individual work	Description here
9	Independent learning ability	Description here

Appendix B

Addenda

B.1 Ethics Forms